Problem Set #61

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Problem 7

(a) The two curves intersect at $\theta = -\pi/3$ and $\theta = \pi/3$.

$$A = \frac{1}{2}\int_{-\pi/3}^{\pi/3} \left(r_1^2 - r_2^2\right) \mathrm{d}\theta = \frac{1}{2}\int_{-\pi/3}^{\pi/3} \left(4\cos^2\theta - 1\right) \mathrm{d}\theta = 2\int_{-\pi/3}^{\pi/3} \frac{1+\cos 2\theta}{2} \mathrm{d}\theta - \frac{1}{2}\left[\theta\right]_{-\pi/3}^{\pi/3}$$
$$= \left[\theta + \frac{\sin 2\theta}{2}\right]_{-\pi/3}^{\pi/3} - \frac{1}{2}\left(\frac{\pi}{3} - \left(-\frac{\pi}{3}\right)\right) = \frac{\pi}{3} + \frac{\sqrt{3}}{4} - \left(-\frac{\pi}{3} - \frac{\sqrt{3}}{4}\right) - \frac{1}{2}\frac{2\pi}{3}$$
$$= \frac{\pi}{3} + \frac{\sqrt{3}}{4} + \frac{\pi}{3} + \frac{\sqrt{3}}{4} - \frac{\pi}{3} = \boxed{\frac{\pi}{3} + \frac{\sqrt{3}}{2}}$$

(b) We need to find when the curves $r^2 = 8\cos 2\theta$ and $r = 2$ intersect.

$$8\cos 2\theta = 2^2 \implies \cos 2\theta = \frac{1}{2} \implies 2\theta \in \left\{\frac{\pi}{3} + 2\pi n, -\frac{\pi}{3} + 2\pi n\right\} \implies \theta \in \left\{\frac{\pi}{6} + \pi n, -\frac{\pi}{6} + \pi n\right\}$$

We choose the intersections at $\theta = -\pi/6$ and $\theta = \pi/6$.

$$A = \frac{1}{2}\int_{-\pi/6}^{\pi/6} (8\cos 2\theta - 4) \mathrm{d}\theta = \frac{1}{2}\left[4\sin 2\theta - 4\theta\right]_{-\pi/6}^{\pi/6} = \frac{1}{2}\left(4\cdot\frac{\sqrt{3}}{2} - \frac{2\pi}{3} - \left(-4\cdot\frac{\sqrt{3}}{2} - 4\left(-\frac{\pi}{6}\right)\right)\right)$$
$$= \frac{1}{2}\left(2\sqrt{3} - \frac{2\pi}{3} + 2\sqrt{3} - \frac{2\pi}{3}\right) = \frac{1}{2}\left(4\sqrt{3} - \frac{4\pi}{3}\right) = \boxed{2\sqrt{3} - \frac{2\pi}{3}}$$

(c) $3\cos\theta = 1 + \cos\theta \implies \cos\theta = \frac{1}{2} \implies \theta \in \left\{\frac{\pi}{3} + 2\pi n, -\frac{\pi}{3} + 2\pi n\right\}$

$$A = \frac{1}{2}\int_{-\pi/3}^{\pi/3} \left((3\cos\theta)^2 - (1+\cos\theta)^2\right) \mathrm{d}\theta = \frac{1}{2}\int_{-\pi/3}^{\pi/3} \left(9\cos^2\theta - 1 - 2\cos\theta - \cos^2\theta\right) \mathrm{d}\theta$$
$$= \frac{1}{2}\int_{-\pi/3}^{\pi/3} \left(8\cos^2\theta - 2\cos\theta - 1\right) \mathrm{d}\theta = 4\int_{-\pi/3}^{\pi/3} \frac{1+\cos 2\theta}{2} \mathrm{d}\theta - \frac{1}{2}\left[2\sin\theta + \theta\right]_{-\pi/3}^{\pi/3}$$
$$= 2\left[\theta + \frac{\sin 2\theta}{2}\right]_{-\pi/3}^{\pi/3} - \frac{1}{2}\left(\sqrt{3} + \frac{\pi}{3} + \sqrt{3} + \frac{\pi}{3}\right) = 2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{4} + \frac{\pi}{3} + \frac{\sqrt{3}}{4}\right) - \left(\sqrt{3} + \frac{\pi}{3}\right)$$
$$= 2\left(\frac{2\pi}{3} + \frac{\sqrt{3}}{2}\right) - \sqrt{3} - \frac{\pi}{3} = \frac{4\pi}{3} + \sqrt{3} - \sqrt{3} - \frac{\pi}{3} = \frac{3\pi}{3} = \boxed{\pi}$$

Problem 8

(a) Intersection is at $\theta = \pi/3$.

$$A = \frac{1}{2}\int_0^{\pi/3} \sin^2\theta \,\mathrm{d}\theta + \frac{1}{2}\int_{\pi/3}^{\pi/2} 3\cos^2\theta \,\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi/3} \frac{1-\cos 2\theta}{2} \mathrm{d}\theta + \frac{3}{2}\int_{\pi/3}^{\pi/2} \frac{1+\cos 2\theta}{2} \mathrm{d}\theta$$
$$= \frac{1}{4}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/3} + \frac{3}{4}\left[\theta + \frac{\sin 2\theta}{2}\right]_{\pi/3}^{\pi/2} = \frac{1}{4}\left(\frac{\pi}{3} - \frac{\sqrt{3}}{4}\right) + \frac{3}{4}\left(\frac{\pi}{2} + 0 - \frac{\pi}{3} - \frac{\sqrt{3}}{4}\right)$$
$$= \frac{\pi}{12} - \frac{\sqrt{3}}{16} + \frac{3}{4}\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right) = \frac{\pi}{12} - \frac{\sqrt{3}}{16} + \frac{\pi}{8} - \frac{3\sqrt{3}}{16} = \frac{2\pi}{24} + \frac{3\pi}{24} - \frac{4\sqrt{3}}{16} = \boxed{\frac{5\pi}{24} - \frac{\sqrt{3}}{4}}$$

(b) There is one intersection at $\theta = (\pi/4)/2 = \pi/8$. This is a rose curve with 4 petals, and $4 \times 2 = 8$ areas of intersection. Let A_1 be the area of one such intersection.

$$A_1 = \frac{1}{2}\int_0^{\pi/8} \sin^2 2\theta \,\mathrm{d}\theta + \frac{1}{2}\int_{\pi/8}^{\pi/4} \cos^2\theta \,\mathrm{d}\theta = \frac{1}{2}\int_0^{\pi/8} \frac{1-\cos 4\theta}{2} \mathrm{d}\theta + \frac{1}{2}\int_{\pi/8}^{\pi/4} \frac{1+\cos 4\theta}{2} \mathrm{d}\theta$$
$$= \frac{1}{4}\left[\theta - \frac{\sin 4\theta}{4}\right]_0^{\pi/8} + \frac{1}{4}\left[\theta + \frac{\sin 4\theta}{4}\right]_{\pi/8}^{\pi/4} = \frac{1}{4}\left(\frac{\pi}{8} - \frac{1}{4}\right) + \frac{1}{4}\left(\frac{\pi}{4} + 0 - \frac{\pi}{8} - \frac{1}{4}\right)$$
$$= \frac{\pi}{32} - \frac{1}{16} + \frac{\pi}{16} - \frac{\pi}{32} - \frac{1}{16} = \frac{\pi}{16} - \frac{1}{8}$$
$$A = 8A_1 = 8\left(\frac{\pi}{16} - \frac{1}{8}\right) = \boxed{\frac{\pi}{2} - 1}$$